

- Burns, D. P., and R. H. Davidson. 1966. The amino acids and sugars in honeydew of the tuliptree scale, *Toumeyella liriodendri* (Gmelin), and in the sap of its host, yellow-poplar. *Ann. Entomol. Soc. Amer.* 59: 1071-3.
- Ewart, W. H., and R. L. Metcalf. 1956. Preliminary studies of sugars and amino acids in the honeydews of five species of coccids feeding on citrus in California. *Ibid.* 49: 441-7.
- Mittler, T. E. 1958. Studies on the feeding and nutrition of *Tuberolachnus salignus*. III. The nitrogen economy. *J. Exp. Biol.* 35: 626-38.

Temperature Limits of Flight of the Mountain Pine Beetle, *Dendroctonus ponderosae*^{1,2}

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Minimum temperatures required for flight of many species of *Dendroctonus* have been determined (Atkins 1966, McMullen and Atkins 1962, Miller and Keen 1960, Shepherd 1966, and others). Most of the investigators report flight thresholds between 16 and 21°C (61 and 70°F), several at ca. 20°C.

The upper limits are less well known. Rudinsky and Vité (1956) reported *D. pseudotsugae* Hopkins become restless and suffer thermal paralysis at temperatures between 33 and 36°C (91 and 97°F). Shepherd (1966) reported *D. ponderosae* Hopkins become negatively phototactic when temperatures exceed 35°C. Gara and Vité (1962) place the upper limit for *D. brevicornis* Hopkins flight between 33 and 35°C.

This study summarizes laboratory data on the temperature range within which the mountain pine beetle can fly, with particular emphasis placed on the extremes.

MATERIALS AND METHODS

The flying ability of mountain pine beetles at various temperatures was determined in the laboratory during

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winter and early spring months of 1968 and 1969. Test beetles were reared from bolts cut near Redfeather Lakes, Colo., from field-infested ponderosa pines, *Pinus ponderosa* Laws. Only strong flyers were used. Beetles selected flew at least twice to a window 10 ft from the release point in a room at 73±1°F, and 50% RH. Lots of 10 nonsexed, strong flyers were then subjected to a pre-conditioning period of either 0.5, 1.0, 2.0, or 4.0 hr at the temperature at which they would be tested for flight: 55, 60, 65, 70, 90, 95, 100, or 105°F. Relative humidity was held at 50%. Four lots of 10 beetles each were tested at each of 75, 80, and 85°F, but only at the 0.5-hr conditioning period, because I was not concerned with flight at these more optimum temperatures. During the pre-conditioning period, the beetles were held individually in gelatin capsules in total darkness.

Conditioned beetles were released in groups of 10 on a small release platform in a constant temperature-humidity cabinet (27 ft³) held at the same temperature and relative humidity as the conditioning cabinet. An incandescent light outside the cabinet's single glass window produced 750 lux at the release platform. The number of beetles that flew 6 or more inches was tallied during each 5-min test period. Beetles were tested once and discarded.

RESULTS

Essentially no mountain pine beetles flew until the temperature exceeded 65°F (Table 1). The single flyer recorded at that temperature probably was inadequately conditioned. The indicated threshold for flight was ca. 67-68°F.

Good flight response was evident at 90 and 95°F throughout all conditioning periods tested. At 100°F, flight response was good if beetles had not been held at that temperature for more than 1 hr. Longer exposures weakened response noticeably. Flight at temperatures above 100°F was severely restricted, and was essentially nonexistent at 105°F if exposure lasted much beyond 1 hr. The maximum air temperature at which the beetle will fly, in 50% RH and not exposed to direct sunlight, appears from extrapolation of the data to be ca. 107°F.

Flight has been observed in the field at temperatures below those reported here. This flight resulted from the warming influence of sunlight, either directly on emerged beetles, or on the bark in the immediate vicinity of beetles

Table 1.—Percent of mountain pine beetles that flew 6 or more inches at various temperatures and pretest conditioning periods.

Test temp (°F)	Beetles conditioned for:							
	0.5 hr		1.0 hr		2.0 hr		4.0 hr	
	No. tested	% flew	No. tested	% flew	No. tested	% flew	No. tested	% flew
55	50	0	..	..	..	..	..	..
60	40	0	10	0	..	..	..	..
65	60	2	20	0	20	0	20	0
70	70	21	60	28	60	8	60	13
75	40	60	..	..	..	..	..	..
80	40	75	..	..	..	..	..	..
85	40	57	..	..	..	..	..	..
90	40	57	50	62	50	66	50	60
95	50	58	30	43	30	47	30	53
100	30	60	40	62	30	33	40	7
105	50	20	40	15	40	0	..	..

capable of flight. This phenomenon has previously been observed by numerous workers investigating flight behavior of bark beetles.

The data presented here should help to better understand flight requirements of mountain pine beetles and perhaps contribute to more efficient use of natural and synthetic attractant materials.

REFERENCES CITED

- Atkins, M. D. 1966. Laboratory studies on the behavior of the Douglas-fir beetle, *Dendroctonus pseudotsugae* Hopkins. Can. Entomol. 98: 953-91.
- Gara, R. I., and J. P. Vité. 1962. Studies on the flight patterns of bark beetles (Coleoptera: Scolytidae) in second growth ponderosa pine forests. Contrib. Boyce Thompson Inst. Plant Res. 21: 275-90.
- McMullen, L. H., and M. D. Atkins. 1962. On the flight and host selection of the Douglas-fir beetle, *Dendroctonus pseudotsugae* Hopk. (Coleoptera: Scolytidae). Can. Entomol. 94: 1309-25.
- Miller, J. M., and F. P. Keen. 1960. Biology and control of the western pine beetle. USDA Misc. Publ. 800. 381 p.
- Rudinsky, J. A., and J. P. Vité. 1956. Effects of temperature upon the activity and the behavior of the Douglas fir beetle. Forest Sci. 2: 258-67.
- Shepherd, R. F. 1966. Factors influencing the orientation and rates of activity of *Dendroctonus ponderosae* Hopkins (Coleoptera: Scolytidae). Can. Entomol. 98: 507-18.
- nature, so a need existed for additional techniques that could be used in the study of these organs. Whiting (1950) reported a method of preparing whole mounts of *Arbacia* eggs, *Chaetopterus* eggs and larvae, small nematodes, small insect ovaries, eggs, and embryos. By modifying this procedure, I was able to obtain consistently good whole-mount preparations from the gonads of adult male and female house flies.
- Gonads of adult house flies were dissected out in a modified Ringer's solution (Morgan and LaBrecque 1964), transferred to miniature holding containers, and placed in Kahle's fixative (Whiting 1950) for 12-24 hr. The tissue was hydrolized with normal hydrochloric acid, stained with Schiff's reagent (McManus and Mowry 1960), destained with a sulfurous acid rinse (McManus and Mowry 1960), dehydrated, and cleared with triethyl phosphate and xylene (Whiting 1950) by the following schedule:

1. Kahle's fixative, 12-24 hr
2. Distilled water, 2 changes, 1 hr each
3. Normal HCL, 26°C, 10 min
4. Normal HCL, 60°C, 10 min
5. Schiff's reagent, 2 hr
6. Sulfurous acid rinse, 10 min
7. Distilled water, 2 changes, 10 min each
8. Triethyl phosphate, 3 changes, 15 min each
9. Triethyl phosphate and xylene (1:1), 2 changes, 15 min each
10. Xylene, 2 changes, 24 hr each.

The testes and ovaries were removed from the xylene and transferred to a drop of thick mounting medium. The entire ovary was either left intact, or the ovarioles were gently teased apart before the coverslip was added. Consistently good whole-mount preparations of the gonads of male and female house flies resulted.

A Procedure for Staining Whole Mounts of Gonads of House Flies^{1,2}

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The adult house fly, *Musca domestica* L., has proved an excellent experimental animal for studies of chemosterilants and their effects on reproductive tissue. However, the majority of this research is cytological in

REFERENCES CITED

- McManus, J. F. A., and R. W. Mowry. 1960. Staining Methods: Histological and Histochemical. Paul B. Hoeber, Inc., New York. 423 p.
- Morgan, P. B., and G. C. LaBrecque. 1964. Preparation of house fly chromosomes. Ann. Entomol. Soc. Amer. 57(2): 794-5.
- Whiting, A. R. 1950. A modification of the Schmuck-Metz whole mount techniques for chromosome study. Stain Technol. 25: 21-22.

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